

Useful Memory under Drift

Temporal-Validity Horizons for Finite-Memory Distribution Tracking

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Abstract

We investigate the fundamental limits of tracking an evolving distribution under temporal drift. Longer memory reduces finite-sample estimation noise but also introduces staleness through age mismatch with the present state. This balance is formalized by a finite temporal-validity horizon, which induces a near-optimal useful-memory region around the scale $n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}$; the finite-sample rate exponent a and the temporal roughness exponent H govern the expansion and contraction of that region. These theoretical limits are established within a tractable one-dimensional framework, with a Gaussian lower bound showing that the boundary is structural rather than estimator-specific. Finally, empirical evidence confirms the predicted interior memory optima, roughness-dependent scaling, the same retention signature on a real stream, and a measurable lag between the loss of temporal validity and the activation of standard changepoint alarms.

Keywords: tracking under drift; temporal validity; finite-memory tracking; distribution tracking; Wasserstein distance; Sinkhorn divergence.

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1 Introduction

In stationary statistics, memory is an unambiguous resource: larger samples reduce variance because the target distribution does not change. Under drift, that logic fails. In streaming estimation, monitoring, and online adaptation, the evidence retained in memory was generated by distributions that may no longer match the current state. More memory therefore does two things at once: it reduces finite-sample noise, and it increases the age of the evidence used to estimate the current state.

That tension creates a horizon problem. Short windows are fresh but noisy. Long windows are stable but stale. The temporal-validity horizon asks how long retained evidence remains useful for estimating the present before drift-induced misalignment overtakes the remaining variance gain.

A general horizon law captures this problem. The law takes the form

$$\mathbb{E} d(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-a} + C_S \zeta n^H,$$

where $a > 0$ is the finite-sample rate exponent of the chosen geometry and estimator, $H \in (0, 1]$ is the temporal roughness exponent of the drift path, and ζ is the roughness amplitude. Balancing these terms yields the temporal-validity horizon

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The horizon depends jointly on finite-sample geometry and on the temporal regularity with which drift accumulates staleness.

At any fixed tolerance $\delta > 0$, the same envelope induces a useful-memory region: the memory lengths whose total tracking error remains within $(1 + \delta)$ times the optimum. The horizon fixes the scale of that region, and its normalized shape depends only on the exponents a and H .

The tractable one-dimensional bounded-support fixed-span triangular-array model establishes the law through the quantile representation of W_2^2 , a triangular-array Bahadur expansion, and a design-effect comparison. In that regime, the root- n finite-sample rate

$$\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1/2}).$$

under explicit compact-support, fixed-span, and integrated-remainder hypotheses. This establishes the horizon law in a tractable one-dimensional bounded-support fixed-span model.

The structural lower bound shows that the horizon is imposed by the drift geometry itself. A Gaussian lower-bound construction sharpens the constant-level geometry, and in the Gaussian location model the same object yields a full minimax benchmark on deterministic H -older paths.

For fixed- ε Sinkhorn geometry on an embedded fixed-span model, the proved structural ingredients yield pointwise and bandwise one-dimensional fixed- ε inheritance results on compact support, while the embedded low-dimensional extension isolates the remaining operational frontier.

Broader regular distribution families and operational geometries support different finite-sample rates, and therefore different members of the horizon family. Memory under drift is governed by the interaction between finite-sample geometry and temporal roughness.

The general horizon law, the useful-memory region, the uniform-window staleness control, the one-dimensional proof-model upper bound, the Gaussian structural lower bound, and the Gaussian location benchmark are

theorem-level results. The fixed- ε Sinkhorn result set includes the structural ingredients together with the point-wise and bandwise one-dimensional fixed- ε inheritance theorems. The numerical evidence concentrates on moderate compact bands bounded away from zero, where the same horizon logic appears operationally stable.

Useful memory is the near-optimal finite-memory region induced by the temporal-validity horizon. It is the set of memory lengths whose total tracking error remains within a fixed multiplicative tolerance of the optimum.

The finite-memory U-curve, roughness-dependent horizon scaling, the fixed- ε Sinkhorn evidence, and the measurable lag between validity loss and the first detector alarm are visible in the experiments. Temporal validity and changepoint evidence are different statistical objects: retained evidence can stop representing the present before change becomes detectable. The experiments are synthetic and structurally targeted; they reveal the predicted signatures without attempting a benchmark comparison against adaptive tracking baselines.

Adjacent literatures include dynamic regret, adaptive windowing, locally stationary smoothing, adaptive filtering, and distributional tracking under transport geometry. Dynamic-regret analysis asks how well one tracks a moving comparator, and adaptive-window or drift-detection methods ask when the accumulated evidence is sufficient to react. For finite-memory tracking under drift, the central object is the time scale on which retained evidence remains temporally valid for estimating the present distribution before age becomes the dominant source of error.

2 Useful Memory and Temporal-Validity Horizons

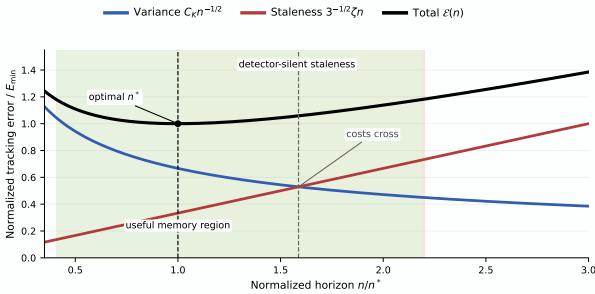


Figure 1: Finite-sample–staleness trade-off under drift. The finite-sample term falls with horizon, the staleness term rises with age, and total error is minimized at the temporal-validity horizon n^* . The green band marks a near-optimal useful-memory region, and the red band marks pre-detection validity loss: retained evidence can stop representing the present before enough changepoint evidence accumulates to trigger an alarm.

For a finite-memory estimator with lag weights $w_0, \dots, w_{n-1}$, the natural age variable is the effective lag

$$A(w) := \sum_{j=0}^{n-1} j w_j.$$

Its corresponding staleness at time t is the weighted transport misalignment

$$S_t(w) := \sum_{j=0}^{n-1} w_j W_2(P_{t-j}, P_t).$$

For a uniform window, $A(w) = (n-1)/2$, so window length and temporal age are the same object up to constants.

Throughout, P_t denotes the present target law, $\bar{P}_t^{(n)}$ the window target obtained by averaging the last n target laws, and $\hat{P}_t^{(n)}$ the estimator built from the last n observations. The finite-sample term always refers to the estimation error relative to $\bar{P}_t^{(n)}$, while the staleness term refers to the discrepancy between $\bar{P}_t^{(n)}$ and P_t .

The theory combines three ingredients: a finite-sample bound relative to $\bar{P}_t^{(n)}$, a path-class control on the staleness between $\bar{P}_t^{(n)}$ and P_t , and, when available, a lower-bound or benchmark theorem showing that the resulting horizon is structural on the geometry under study. The horizon scale then induces a useful-memory region: the range of memory lengths whose tracking error remains near optimal.

2.1 General Horizon Law

The main theorem separates finite-sample behavior from drift-induced staleness.

Theorem 2.1 (Abstract upper law). *Let d be a probability metric satisfying the triangle inequality. At time t , let P_t be the present target law, let $\bar{P}_t^{(n)}$ be the window target, and let $\hat{P}_t^{(n)}$ be the estimator built from the last n observations. Assume that for some constants $C_K > 0$, $C_S > 0$, exponent $a > 0$, roughness index $H \in (0, 1]$, and amplitude $\zeta > 0$,*

$$\mathbb{E} d(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \leq C_K n^{-a}$$

and

$$d(\bar{P}_t^{(n)}, P_t) \leq C_S \zeta n^H$$

hold for all n in the regime of interest. Then

$$\mathbb{E} d(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-a} + C_S \zeta n^H.$$

Proof. By the triangle inequality,

$$d(\hat{P}_t^{(n)}, P_t) \leq d(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) + d(\bar{P}_t^{(n)}, P_t).$$

Taking expectations and applying the two assumptions yields the claim. $\square$

The theorem isolates the structure of the problem: the path class supplies the staleness term, while the chosen geometry and sample regime supply the finite-sample term.

Corollary 2.2 (Temporal-validity horizon). *Under the assumptions of theorem 2.1, the upper envelope*

$$\Phi(n) := C_K n^{-a} + C_S \zeta n^H$$

is minimized, up to constant factors depending only on a , H , and C_S , at

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)},$$

and the corresponding error scale is

$$\Phi(n^*) \asymp C_K^{H/(a+H)} \zeta^{a/(a+H)}.$$

Proof. Balance the decreasing finite-sample term against the increasing staleness term:

$$C_K n^{-a} \asymp C_S \zeta n^H.$$

This gives $n^{a+H} \asymp C_K/\zeta$, hence the displayed horizon law. Substituting that scale back into either term gives the optimal error scale. $\square$

Write the continuous optimizer of Φ as

$$n_\star := \left(\frac{aC_K}{HC_S\zeta} \right)^{1/(a+H)}.$$

This optimizer differs from the scale $n^*(a, H)$ only by constants depending on a , H , and C_S , so the asymptotic horizon law remains unchanged.

2.2 Useful-Memory Region

The horizon scale induces a near-optimal region of memory lengths. For any $\delta > 0$, define the useful-memory region by

$$\mathcal{U}_\delta := \{n > 0 : \Phi(n) \leq (1 + \delta)\Phi(n_\star)\}.$$

This object records not only where the envelope is minimized, but also how much memory remains acceptable up to a fixed tolerance.

Proposition 2.3 (Useful-memory region at tolerance δ). *Let $\Phi(n) = C_K n^{-a} + C_S \zeta n^H$ with $a, H > 0$, and let $n_\star$ be its continuous optimizer. Then the normalized profile*

$$\Psi(x) := \frac{\Phi(n_\star x)}{\Phi(n_\star)} = \frac{Hx^{-a} + ax^H}{a + H}$$

is strictly decreasing on $(0, 1]$ and strictly increasing on $[1, \infty)$. Consequently,

$$\mathcal{U}_\delta = [n_-, n_+] = n_\star [x_-(\delta; a, H), x_+(\delta; a, H)],$$

where $x_-(\delta; a, H) \leq 1 \leq x_+(\delta; a, H)$ are the two solutions of $\Psi(x) = 1 + \delta$. In particular, the useful-memory region scales linearly with the horizon scale $n_\star$, and its relative shape depends only on a , H , and δ .

Proof. Since $n_\star$ minimizes Φ , differentiation gives

$$aC_K n_\star^{-a} = HC_S \zeta n_\star^H.$$

Substituting $n = n_\star x$ into Φ and dividing by $\Phi(n_\star)$ yields the displayed identity for Ψ . Differentiation then gives

$$\Psi'(x) = \frac{aH}{a+H} (x^{H-1} - x^{-a-1}),$$

which is negative for $x < 1$ and positive for $x > 1$. Since $\Psi(1) = 1$ and $\Psi(x) \rightarrow \infty$ as $x \rightarrow 0$ or $x \rightarrow \infty$, the sublevel set $\{x > 0 : \Psi(x) \leq 1 + \delta\}$ is a compact interval containing 1, which proves the representation of $\mathcal{U}_\delta$. $\square$

For small tolerance, the useful-memory region has the local expansion

$$x_\pm(\delta; a, H) = 1 \pm \sqrt{\frac{2\delta}{aH}} + o(\sqrt{\delta}), \quad \delta \downarrow 0,$$

because $\Psi''(1) = aH$. The local curvature therefore controls how quickly the near-optimal region contracts around the horizon.

The horizon depends jointly on the finite-sample rate exponent and the roughness exponent. Different geometries and different drift regularities induce different memory scales.

2.3 Drift Regularity and Staleness Accumulation

The staleness side of the law is controlled by a deterministic path class. For $H \in (0, 1]$ and $\zeta > 0$, write

$$\mathfrak{P}_H(\zeta) := \{(P_s)_{s \in \mathbb{Z}} : W_2(P_{t-j}, P_t) \leq \zeta j^H \text{ for all } j \geq 0\}.$$

The case $H = 1$ is the local Lipschitz envelope; smaller H correspond to slower accumulation of staleness with lag.

Lemma 2.4 (Hölder staleness bound via averaged couplings). *Let*

$$\bar{P}_t^{(n)} := \frac{1}{n} \sum_{j=0}^{n-1} P_{t-j}.$$

If the target path belongs to $\mathfrak{P}_H(\zeta)$, then

$$W_2^2(\bar{P}_t^{(n)}, P_t) \leq \frac{\zeta^2}{n} \sum_{j=0}^{n-1} j^{2H},$$

and therefore, by Jensen's inequality,

$$W_2(\bar{P}_t^{(n)}, P_t) \leq \zeta \left(\frac{1}{n} \sum_{j=0}^{n-1} j^{2H} \right)^{1/2} = C_{H,n} \zeta n^H,$$

where

$$C_{H,n} := \left(n^{-(2H+1)} \sum_{j=0}^{n-1} j^{2H} \right)^{1/2}.$$

In particular, $C_{H,n} \rightarrow (2H+1)^{-1/2}$ as $n \rightarrow \infty$.

Proof. For each lag j , let π_j be an optimal coupling between P_{t-j} and P_t . Their average

$$\bar{\pi} := \frac{1}{n} \sum_{j=0}^{n-1} \pi_j$$

couples $\bar{P}_t^{(n)}$ with P_t , so

$$\begin{aligned} W_2^2(\bar{P}_t^{(n)}, P_t) &\leq \int \|x - y\|^2 d\bar{\pi}(x, y) \\ &= \frac{1}{n} \sum_{j=0}^{n-1} W_2^2(P_{t-j}, P_t). \end{aligned}$$

Applying the Hölder envelope gives the displayed finite- n constant, and taking square roots completes the proof. $\square$

Combining lemma 2.4 with corollary 2.2 yields the horizon family

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

In the root- n finite-sample regime $a = 1/2$,

$$n^*(1/2, H) \asymp (C_K/\zeta)^{2/(1+2H)}.$$

The finite- n constant in lemma 2.4 is the constant used here for uniform windows; the simpler $(2H + 1)^{-1/2}$ expression is its large-window limit.

2.4 One-Dimensional Proof Model for the Root- n Regime

The bounded-support fixed-span one-dimensional triangular array provides a tractable rigorous finite-sample instantiation of the abstract law. The model is built on common compact support, uniformly regular window mixtures, and an integrated Bahadur remainder hypothesis.

Proposition 2.5 (Root- n finite-sample bound in the 1D proof model). *Let $X_{n,1}, \dots, X_{n,n}$ be independent one-dimensional observations with laws $P_{n,1}, \dots, P_{n,n}$, and define the window mixture*

$$\bar{P}_n = \frac{1}{n} \sum_{j=1}^n P_{n,j}.$$

Assume:

- (i) all $P_{n,j}$ are supported on a common compact interval $[L, U]$;
- (ii) the within-window drift span is uniformly bounded in n ;
- (iii) each mixture law $\bar{P}_n$ has a density $\bar{f}_n$ on $[L, U]$ that is bounded below by a positive constant and uniformly Hölder;
- (iv) writing q_n for the quantile function of $\bar{P}_n$ and $\hat{q}_n$ for the empirical quantile function of $\hat{P}_n^{\text{tri}}$, one has the Bahadur representation

$$\hat{q}_n(u) - q_n(u) = \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u),$$

$$u \in (0, 1),$$

with

$$\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1}).$$

Then

$$\mathbb{E} W_2^2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1}),$$

and therefore, by Jensen's inequality,

$$\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1/2}).$$

Proof. In one dimension,

$$W_2^2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = \int_0^1 (\hat{q}_n(u) - q_n(u))^2 du,$$

so the problem reduces to the integrated L_2 error of the empirical quantile process. By item (iv),

$$\hat{q}_n(u) - q_n(u) = \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u).$$

Define

$$Z_n(u) := \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))}.$$

Then

$$\mathbb{E} \int_0^1 (\hat{q}_n(u) - q_n(u))^2 du = \mathbb{E} \int_0^1 (Z_n(u) + R_n(u))^2 du.$$

Let $m > 0$ be a uniform lower bound for $\bar{f}_n$. Since

$$\hat{F}_n(x) = \frac{1}{n} \sum_{j=1}^n \mathbf{1}\{X_{n,j} \leq x\},$$

independence across the row gives

$$\text{Var}(\hat{F}_n(q_n(u))) = \frac{1}{n^2} \sum_{j=1}^n p_{n,j}(u)(1 - p_{n,j}(u)) \leq \frac{1}{4n},$$

where $p_{n,j}(u) = P_{n,j}((-\infty, q_n(u)])$. Therefore

$$\mathbb{E} \int_0^1 Z_n(u)^2 du \leq m^{-2} \int_0^1 \text{Var}(\hat{F}_n(q_n(u))) du \leq \frac{1}{4m^2 n}.$$

By item (iv),

$$\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1}).$$

The cross term is then $o(n^{-1})$ by Cauchy–Schwarz, since $\mathbb{E} \int_0^1 Z_n(u)^2 du = O(n^{-1})$ and $\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1})$. Hence the expected squared Wasserstein error is $O(n^{-1})$, and Jensen's inequality gives the root- n rate for $\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n)$.

The main conclusion is the exponent. Under this bounded-support fixed-span regime, the triangular-array finite-sample term remains root- n . $\square$

Remark 2.6 (Sufficient conditions for the proof model). The substantive assumption in proposition 2.5 is item (iv). For i.i.d. quantile processes, Bahadur representations of this type are classical; see Bahadur [1966] and van der Vaart [1998, Chapter 21]. In the fixed-span triangular-array model, a concrete sufficient package is a common compact support $[L, U]$; rowwise densities $f_{n,j}$ uniformly bounded above and below by positive constants; a uniform smoothness bound on $f_{n,j}$ across rows; and a rowwise quantile process whose integrated Bahadur remainder satisfies $\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1})$. This package is compatible with bounded smooth monotone deformations of a common base law with uniformly bounded derivatives, and with fixed-support location perturbations when the support constraint is preserved. Separate diagnostics on a smooth fixed-support deformation model show integrated residual decay faster than n^{-1} across the tested span and roughness values, which is consistent with item (iv) on that class.

2.5 Structural Lower Bound in the Root- n Regime

The upper law would be substantially weaker if it were only an estimator-specific statement. The lower bound shows that, in the root- n finite-sample regime $a = 1/2$, the horizon is imposed by the drift geometry itself.

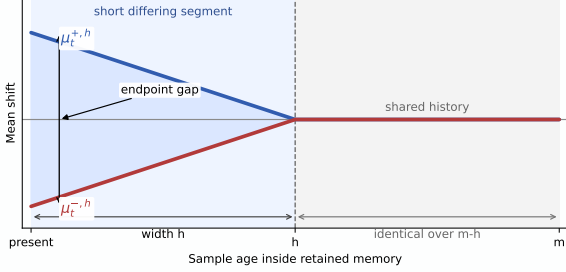


Figure 2: Gaussian lower-bound construction. The two paths share most of the retained history and differ only on a recent segment of width h , yet remain separated at the present. That recent segment already produces the same horizon exponent as the upper law in the root- n regime.

Proposition 2.7 (Structural lower bound in the root- n regime). *Fix $H \in (0, 1]$ and work on the Gaussian location subclass with known variance σ^2 and path pair*

$$\mu_{t-j}^{\pm,h} = \pm\beta(h-j)_+^H, \quad j = 0, 1, \dots, h,$$

where $\beta \leq \zeta$ enforces the roughness budget. Let $\mathcal{G}_{H,\zeta}^{\text{Gauss}}$ denote the corresponding Gaussian location subclass of target paths, and suppose the retained observations are independent with laws $N(\mu_{t-j}^{\pm,h}, \sigma^2)$. Then there exists a constant $c_H > 0$ depending only on H such that the endpoint tracking risk over this subclass satisfies

$$\inf_{\hat{P}} \sup_{P \in \mathcal{G}_{H,\zeta}^{\text{Gauss}}} \mathbb{E} W_2(\hat{P}_t, P_t) \geq c_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Consequently, in the root- n finite-sample regime $a = 1/2$, the horizon scale

$$h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$$

is structural even on this restricted subclass.

Proof outline. Compare two hypotheses generated by the path pair $\mu_{t-j}^{+,h}$ and $\mu_{t-j}^{-,h}$. At the present time, the endpoint separation is of order βh^H . On the Gaussian location subclass, this is also the W_2 separation. The Kullback–Leibler divergence between the two hypotheses is proportional to

$$\beta^2 \sum_{r=1}^h r^{2H} / \sigma^2.$$

Choosing

$$\beta = \min \left\{ \zeta, \frac{\sigma}{2\sqrt{\sum_{r=1}^h r^{2H}}} \right\}$$

keeps the testing problem in the Le Cam regime while respecting the roughness budget. The resulting two-point lower bound is of order βh^H . Balancing the active

roughness constraint against the testing constraint gives $h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$ and therefore the displayed lower law. The exact Gaussian two-point calculation used here is recorded in appendix A.

The lower bound is subclass based and not class-tight, but it is enough to show that the horizon is a structural feature of the problem and not an estimator artifact.

The proposition above is the lower-bound fact needed for the main horizon law. The same Gaussian construction also exposes the constant-level geometry. The next proposition records the sharp Gaussian constants available within that proof scheme.

2.6 Gaussian Lower-Bound Geometry

Let Φ and φ denote the standard normal distribution function and density.

Proposition 2.8 (Gaussian lower-bound constants). *Fix $H \in (0, 1]$ and set $p_H := 2H/(2H+1)$. Let $x_H > 0$ be the unique maximizer of $x^{p_H} \Phi(-x)$, equivalently the unique solution of*

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

For the Gaussian ramp profile $\mu_{t-j}^{\pm,h} = \pm\zeta(h-j)_+^H$, let

$$L_h^{\text{ramp}}(\sigma, \zeta) := \zeta h^H \Phi \left(-\frac{\zeta}{\sigma} \sqrt{\sum_{r=1}^h r^{2H}} \right).$$

Then, as $\sigma/\zeta \rightarrow \infty$,

$$\sup_{h \geq 1} L_h^{\text{ramp}}(\sigma, \zeta) \sim C_H^{\text{ramp}} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)},$$

with

$$C_H^{\text{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Among all endpoint-saturating Hölder profiles, the profile

$$g_r^{\min} = h^H - (h-r)^H$$

minimizes the testing energy and has asymptotic energy constant

$$I_H = \int_0^1 (1 - (1-u)^H)^2 du = \frac{2H^2}{(H+1)(2H+1)}.$$

If

$$L_h^{\min}(\sigma, \zeta) := \zeta h^H \Phi \left(-\frac{\zeta}{\sigma} \sqrt{\sum_{r=1}^h (g_r^{\min})^2} \right),$$

then, as $\sigma/\zeta \rightarrow \infty$,

$$\sup_{h \geq 1} L_h^{\min}(\sigma, \zeta) \sim C_H^{\min} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)},$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

For $H < 1$, this strictly improves the ramp constant by the factor

$$\frac{C_H^{\min}}{C_H^{\text{ramp}}} = \left(\frac{H+1}{2H^2} \right)^{H/(2H+1)} > 1,$$

while at $H = 1$ the two profiles coincide.

Proof. The exact Gaussian testing problem reduces the two-point lower bound to the displayed quantities L_h^{ramp} and L_h^{min} . After rescaling by $h = A(\sigma/\zeta)^{2/(2H+1)}$, the ramp energy and endpoint-minimal energy have limits $(2H+1)^{-1}$ and I_H , respectively. The corresponding normalized payoff becomes a scalar multiple of $x^{p_H}\Phi(-x)$, whose unique maximizer is x_H . The endpoint-minimal profile minimizes testing energy among all endpoint-saturating Hölder profiles, so it yields the sharper constant. Full details are given in appendix A. $\square$

These constants sharpen the lower-bound picture without changing the horizon exponents or the status of the main results.

The same lower-bound geometry also yields a full class-level benchmark once the model is specialized to Gaussian location tracking. In that setting, the deterministic Hölder envelope, the finite-memory estimator, and the lower-bound construction all live in the same one-dimensional parameterization, so this model admits a direct minimax rate statement.

Theorem 2.9 (Gaussian location minimax benchmark on the deterministic Hölder class). *Fix $H \in (0, 1]$, $\zeta > 0$, and $\sigma > 0$. Let $\mathfrak{M}_H(\zeta)$ denote the class of deterministic mean paths $(\mu_s)_{s \in \mathbb{Z}}$ such that*

$$|\mu_{t-j} - \mu_t| \leq \zeta j^H \quad \text{for all } j \geq 0.$$

Suppose the observations are independent with

$$Y_{t-j} \sim N(\mu_{t-j}, \sigma^2), \quad j = 0, 1, \dots, n-1,$$

and let the target law at time t be $P_t = N(\mu_t, \sigma^2)$. Define the optimal finite-memory minimax risk

$$\mathcal{R}_H^*(\sigma, \zeta) := \inf_{n \geq 1} \inf_{\hat{P}_t^{(n)}} \sup_{\mu \in \mathfrak{M}_H(\zeta)} \mathbb{E} W_2(\hat{P}_t^{(n)}, P_t).$$

Then there exist constants $0 < c_H \leq C_H < \infty$, depending only on H , such that

$$\begin{aligned} c_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)} &\leq \mathcal{R}_H^*(\sigma, \zeta) \\ &\leq C_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}. \end{aligned}$$

In particular, the deterministic Hölder class has horizon scale

$$n_H^* \asymp (\sigma/\zeta)^{2/(2H+1)}.$$

Moreover, the uniform-window mean estimator has an exact asymptotic upper constant on this benchmark. If $y_H > 0$ solves

$$Hy_H(2\Phi(y_H) - 1) = \varphi(y_H),$$

then its asymptotic horizon shape is

$$A_H^{\text{mean}} = ((H+1)y_H)^{2/(2H+1)}$$

and its asymptotic risk constant is

$$\begin{aligned} U_H^{\text{mean}} &= ((H+1)y_H)^{-1/(2H+1)} \\ &\quad \times \left(2\varphi(y_H) + y_H(2\Phi(y_H) - 1) \right). \end{aligned}$$

Proof. For the upper bound, estimate the present mean by the uniform-window average

$$\hat{\mu}_t^{(n)} := \frac{1}{n} \sum_{j=0}^{n-1} Y_{t-j}$$

and set $\hat{P}_t^{(n)} = N(\hat{\mu}_t^{(n)}, \sigma^2)$. Because equal-variance Gaussian laws satisfy

$$W_2(N(m_1, \sigma^2), N(m_2, \sigma^2)) = |m_1 - m_2|,$$

the risk reduces to mean estimation error. The noise term is $\sigma\sqrt{2/(\pi n)}$, while the deterministic bias is bounded by

$$\frac{\zeta}{n} \sum_{j=0}^{n-1} j^H,$$

so

$$\sup_{\mu \in \mathfrak{M}_H(\zeta)} \mathbb{E} W_2(\hat{P}_t^{(n)}, P_t) \leq \sigma\sqrt{\frac{2}{\pi n}} + \frac{\zeta}{n} \sum_{j=0}^{n-1} j^H.$$

Rescaling by $n = A(\sigma/\zeta)^{2/(2H+1)}$ yields a one-parameter normalized upper envelope. Differentiating its exact Gaussian-location form gives the stated root equation for y_H , and therefore the displayed asymptotic upper constant and horizon shape. In particular, this proves the stated upper rate and horizon scale.

For the lower bound, use the symmetric two-point sub-family

$$\mu_{t-j}^{\pm, h} = \pm \beta(h^H - j^H)_+, \quad j \geq 0,$$

with $\beta \leq \zeta$. This family lies in $\mathfrak{M}_H(\zeta)$ because

$$|\mu_t^{\pm, h} - \mu_{t-j}^{\pm, h}| = \beta \min(h^H, j^H) \leq \zeta j^H.$$

Moreover, for fixed endpoint value βh^H , it minimizes the Gaussian testing energy over all symmetric lag profiles allowed by the present-point Hölder envelope, since each lag coordinate is independently minimized by $\beta(h^H - j^H)_+$. The exact Gaussian two-point calculation from the lower-bound analysis in appendix A therefore applies on the full deterministic Hölder class and yields a lower bound of order

$$\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Minimizing over h gives the same horizon scale $(\sigma/\zeta)^{2/(2H+1)}$. Combining the two sides proves the theorem. $\square$

2.7 Benchmark Theorems and Operational Extensions

The Gaussian benchmark and the fixed- ε Sinkhorn extension place the horizon law in two broader settings: a full path-class benchmark with theorem-level closure, and an operational transport geometry in which the one-dimensional compact-support inheritance theory is already explicit. The linear-Gaussian entropic-OT setting suggests an intermediate recursive regime between the i.i.d. fixed- ε theory and the embedded low-dimensional extension [Kalman, 1960, Jazwinski, 1970, Akyildiz et al., 2024].

Fixed- ε Sinkhorn setting. For fixed- ε debiased Sinkhorn, the central question is whether a practical geometry can supply a stable finite-sample term on the embedded fixed-span model. The target law is

$$\mathbb{E} S_\varepsilon(\hat{P}_{t,\text{tri}}^{(n)}, \bar{P}_t^{(n)}) \leq C_\varepsilon n^{-a_\varepsilon}$$

under bounded support, fixed span, and structural low-dimensionality, where S_ε denotes a fixed- ε Sinkhorn-type geometry and the exponent a_ε matches the corresponding i.i.d. mixture benchmark.

Definition 2.10 (Fixed- ε Sinkhorn setting). Fix a bounded embedded fixed-span model, an intrinsic complexity index k , a dual Hölder smoothness level α , and an operational band $\mathcal{B} \subset (0, \infty)$ of regularization values. We say that this model is in the fixed- ε Sinkhorn setting if:

- (i) the i.i.d. fixed- ε benchmark on the same support class admits a finite-sample bound with exponent a_ε on $\mathcal{B}$;
- (ii) the triangular-window and i.i.d.-mixture supports have identical operational covering complexity on $\mathcal{B}$;
- (iii) the dual class lies in the parametric adaptation region $\alpha > k/2$.

Proposition 2.11 (Structural ingredients for fixed- ε Sinkhorn). *On the bounded embedded fixed-span model with squared-Euclidean cost, the first three items in definition 2.10 are structural: the support-complexity inheritance is exact, and the dual-complexity condition is precisely the LCA threshold $\alpha > k/2$.*

Proposition 2.12 (Pointwise fixed- ε inheritance in the 1D proof model). *Assume the hypotheses of proposition 2.5, let $\hat{P}_n^{\text{iid}}$ be the empirical measure of n i.i.d. samples from $\bar{P}_n$, and fix $\varepsilon > 0$. For the debiased Sinkhorn divergence S_ε with squared-Euclidean cost on the common compact support, one has*

$$\mathbb{E} S_\varepsilon(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1/2}).$$

Proof. Write $V_\varepsilon(\mu, \nu)$ for the regularized OT value so that

$$S_\varepsilon(\mu, \nu) = V_\varepsilon(\mu, \nu) - \frac{1}{2} V_\varepsilon(\mu, \mu) - \frac{1}{2} V_\varepsilon(\nu, \nu).$$

Because all measures are supported on one compact interval, they lie in a common bounded quadratic-moment class. By Eckstein and Nutz [2022, Thm. 3.7(ii), Ex. 3.4], for fixed ε there exists $L_\varepsilon < \infty$ depending only on that class and on the cost such that

$$|V_\varepsilon(\mu, \nu) - V_\varepsilon(\mu', \nu')| \leq L_\varepsilon (W_2(\mu, \mu') + W_2(\nu, \nu')).$$

Applying this bound to the cross term and both self terms gives

$$|S_\varepsilon(\mu, \nu) - S_\varepsilon(\mu', \nu')| \leq 2L_\varepsilon (W_2(\mu, \mu') + W_2(\nu, \nu')).$$

With $\nu = \nu' = \bar{P}_n$,

$$S_\varepsilon(\hat{P}_n^{\text{tri}}, \bar{P}_n) \leq S_\varepsilon(\hat{P}_n^{\text{iid}}, \bar{P}_n) + 2L_\varepsilon W_2(\hat{P}_n^{\text{tri}}, \hat{P}_n^{\text{iid}}).$$

Taking expectations and using the triangle inequality gives

$$\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \hat{P}_n^{\text{iid}}) \leq \mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n) + \mathbb{E} W_2(\hat{P}_n^{\text{iid}}, \bar{P}_n).$$

The first term is $O(n^{-1/2})$ by proposition 2.5. The second is the classical i.i.d. one-dimensional quantile-process rate on the same compact support class [Bahadur, 1966, van der Vaart, 1998], so it is also $O(n^{-1/2})$. Hence

$$\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \hat{P}_n^{\text{iid}}) = O(n^{-1/2}).$$

Finally, under the null comparison with fixed ε and compact support, the i.i.d. debiased Sinkhorn term satisfies

$$\mathbb{E} S_\varepsilon(\hat{P}_n^{\text{iid}}, \bar{P}_n) = O(n^{-1})$$

by del Barrio et al. [2023, Thm. 5.1, after fixed- ε rescaling]; see also the null-limit law in Goldfeld et al. [2024]. Combining the bounds yields the displayed $O(n^{-1/2})$ rate. $\square$

Proposition 2.13 (Bandwise fixed- ε inheritance in the 1D proof model). *Let $\mathcal{B} = [\varepsilon_{\min}, \varepsilon_{\max}] \subset (0, \infty)$ be a compact regularization band. Under the hypotheses of proposition 2.12 on a common compact support class, there exists $C_{\mathcal{B}} < \infty$ such that*

$$\sup_{\varepsilon \in \mathcal{B}} \mathbb{E} S_\varepsilon(\hat{P}_n^{\text{tri}}, \bar{P}_n) \leq C_{\mathcal{B}} n^{-1/2}.$$

Consequently, the induced horizon scale is uniform on $\mathcal{B}$:

$$\sup_{\varepsilon \in \mathcal{B}} n_\varepsilon^*(H) \asymp (C_{\mathcal{B}}/\zeta)^{1/(H+1/2)}.$$

Proof. Repeat the proof of proposition 2.12. The stability constant L_ε from Eckstein and Nutz [2022, Thm. 3.7(ii), Ex. 3.4] is bounded on $\mathcal{B}$ because all measures live in one compact support class and $\varepsilon \geq \varepsilon_{\min} > 0$. The i.i.d. debiased-Sinkhorn benchmark on compact support is also uniform on $\mathcal{B}$ by the fixed- ε asymptotics of Goldfeld et al. [2024] and the compact-support null bounds of del Barrio et al. [2023, Thm. 5.1, after fixed- ε rescaling]. Since the triangular-to-i.i.d. W_2 comparison is independent of ε , the constants in the pointwise proof can be chosen uniformly over $\mathcal{B}$. $\square$

Conjecture 2.14 (Bandwise fixed- ε Sinkhorn horizon inheritance). *Beyond the one-dimensional fixed- ε band result above, assume the fixed- ε Sinkhorn setting from definition 2.10. Then, on bands where the i.i.d. benchmark stays in the parametric Sinkhorn regime, the triangular-array finite-sample term should satisfy*

$$\mathbb{E} S_\varepsilon(\hat{P}_{t,\text{tri}}^{(n)}, \bar{P}_t^{(n)}) \leq C_\varepsilon n^{-1/2}, \quad \varepsilon \in \mathcal{B},$$

and the corresponding horizon scale should satisfy

$$n_\varepsilon^*(H) \asymp (C_\varepsilon/\zeta)^{1/(H+1/2)}.$$

More generally, if the i.i.d. benchmark on $\mathcal{B}$ has exponent a_ε , then the operational horizon law should be

$$n_\varepsilon^*(a_\varepsilon, H) \asymp (C_\varepsilon/\zeta)^{1/(a_\varepsilon+H)}.$$

Evidence. The i.i.d. benchmark comes from existing fixed- ε entropic transport results, including fixed-marginal limit theory for the Sinkhorn divergence and intrinsic-complexity adaptation [Goldfeld et al., 2024, Stromme, 2023, Groppe and Hundrieser, 2024]. In the present embedded fixed-span model, support-complexity inheritance is exact and the dual smoothness threshold reduces to $\alpha > k/2$. Calibration then asks whether triangular and i.i.d. rate estimates remain close across the regularization band. Additional one-dimensional smooth fixed-support diagnostics show that triangular and i.i.d. fitted exponents can agree closely on moderate fixed- ε bands, while direct Jacobian diagnostics concentrate the remaining difficulty on uniform control of the self-coupling derivative blocks across nontrivial bands and higher-dimensional embedded classes.

Regular dominated families remain the natural family-level extension class. Local testing geometry and metric comparability suggest the same horizon logic with a modified finite-sample exponent, whereas support-changing and singular families fall outside that regime.

Once a finite-sample regime and a roughness level are specified, the law yields an explicit memory scale and an optimal error scale. Rougher drift and weaker finite-sample rates both force shorter memory and larger residual error.

3 Online Implications

The horizon law identifies the quantities that any adaptive memory rule must track. A policy that updates memory online needs a finite-sample regime, represented by the exponent a or by an operational exponent a_ε , together with a local roughness description of the recent path through the pair (H, ζ) . Once these objects are available, the structural scale $n_\star$ and the useful-memory region $\mathcal{U}_\delta$ follow directly from the horizon law.

In operational geometries, local support structure enters before roughness estimation. The embedded fixed- ε theory already shows that intrinsic complexity controls which finite-sample regime is available through the threshold $\alpha > k/2$. A local proxy such as TwoNN therefore serves as a regime-selection statistic: it does not estimate the horizon by itself, but it selects the finite-sample law to which the horizon formula should be attached.

The roughness side is a separate statistical task. In deterministic Hölder paths, H and ζ belong to an envelope governing how quickly staleness can accumulate with lag. A direct log-log fit to one short empirical sequence of transport discrepancies is generally unstable, because single-path local motion, finite-sample noise, and short-lag discreteness all distort the envelope. The relevant online object is therefore not a raw local slope, but a stabilized estimate built from block averages, activity proxies, or other aggregated recent discrepancies.

This separation gives the minimal architecture of a horizon-informed adaptive policy. First estimate the local regime that determines the finite-sample term. Then estimate a recent roughness envelope for the staleness term. Finally map these estimates to $n_\star$ or to a tolerance band $\mathcal{U}_\delta$, and use that region as the admissible memory range.

Hysteresis, band projection, or validation scoring can stabilize the resulting controller, but those mechanisms operate downstream of the horizon law itself.

The online problem is therefore structurally well posed. The horizon law supplies the target for adaptation; the remaining task is to estimate its ingredients with guarantees strong enough to keep a data-driven memory rule inside the useful-memory region.

4 Empirical Evidence

These experiments expose the finite-sample signatures of the temporal-validity horizon, the useful-memory region induced by that horizon, and evidence relevant to the fixed- ε Sinkhorn frontier.

The Gaussian U-curve uses 20 Monte Carlo seeds, the roughness-family sweep uses 12, the fixed- ε Sinkhorn fits log-log slopes over sample sizes (24, 48, 96, 160) on the pairs (8, 1), (8, 2), (12, 1), and (12, 2) using 24 seeds, and the validity-detection experiment uses a piecewise-ramped drift stream with 12 seeds, a fixed operating window, and both ADWIN and Page-Hinkley detectors. The real-stream diagnostic uses the ELEC2 stream and a short anchor block on the marginal distribution of `nswprice`. The empirical goal is qualitative and structural: persistence of shape, ordering, and sign across repeated runs, not precise constant estimation for every curve.

The empirical signatures are fivefold. Finite-memory risk exhibits an interior optimum together with a near-optimal useful-memory band. That optimum moves with temporal roughness. The same shape appears on a real stream. The fixed- ε Sinkhorn frontier receives structured empirical support. Temporal validity can fail before changepoint evidence becomes statistically visible.

4.1 A Finite-Memory Optimum

The Gaussian horizon sweep isolates the variance-staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the minimum lies at an interior horizon.

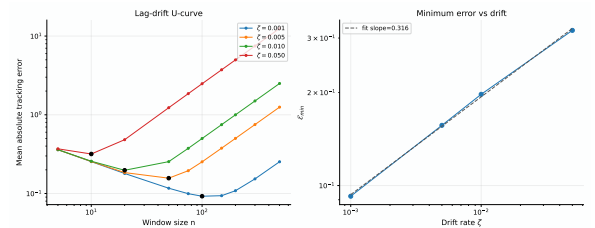


Figure 3: Interior risk optimum over memory length. Tracking error first falls with horizon and then rises once staleness overtakes the remaining variance gain. The optimum remains finite across drift strengths.

This is the basic empirical signature of temporal validity under drift. Under stationarity, more memory would continue to help. Under drift, the same horizon that reduces variance also ages the evidence. Around the minimizing horizon, the curve displays a flat near-optimal region, which is the empirical analogue of the useful-memory region defined from the envelope Φ .

4.2 Roughness-Dependent Horizon Scaling

The next experiment varies the roughness index directly. For each Hölder-type regime, the surrogate error obeys the same structural decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^H,$$

and the experiment records the empirically optimal horizon across a sweep of ζ values.

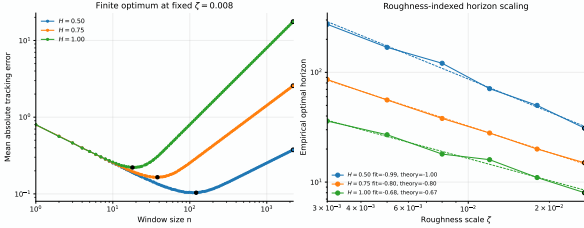


Figure 4: Roughness-dependent horizon scaling. Each path class exhibits a finite optimum, and the location of that optimum changes with H . The horizon is a function of drift regularity, not a fixed window rule.

The empirical optima separate by roughness class, matching the path-dependent horizon logic of corollary 2.2.

4.3 A Real-Stream Diagnostic

The same qualitative signature appears on a real non-stationary stream. Using ELEC2, consider the one-dimensional marginal law of `nswprice`. At each time, approximate the present target by a short anchor block of 48 observations and compare it, in W_1 , with the empirical law induced by the preceding memory window. Averaging this diagnostic across the stream yields a real-data analogue of the finite-memory U-curve.

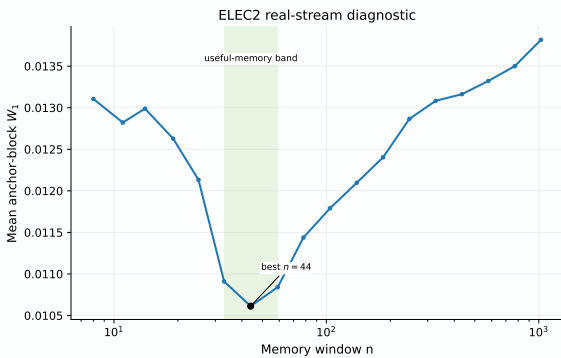


Figure 5: ELEC2 real-stream diagnostic on the `nswprice` marginal. The mean anchor-block W_1 error exhibits a finite interior optimum and a near-optimal useful-memory band. The figure is diagnostic rather than competitive: it asks whether the temporal-validity signature appears on a real stream.

On ELEC2, the diagnostic optimum occurs at a finite memory scale, with a short near-optimal band around it. The real stream therefore exhibits the same basic retention pattern as the synthetic horizon experiments: very short windows are too noisy, while very long windows become stale relative to the present block.

4.4 Operational Sinkhorn Evidence

The fixed- ε Sinkhorn experiments address the bandwise question: when a practical geometry replaces raw W_2 , does the finite-sample term remain stable enough for the horizon law to stay informative in an embedded operational regime? The pointwise and bandwise one-dimensional fixed-support statements appear in propositions 2.12 and 2.13; the calibrated embedded fixed-span model extends the same question to a broader low-dimensional setting.

The fixed- ε Sinkhorn geometry on embedded low-intrinsic-dimensional support shows that the corresponding finite-sample term remains stable across the tested regularization values and that the triangular slope stays close to the i.i.d. benchmark, with no separate exponent emerging. Combined with the structural ingredients isolated in proposition 2.11, this identifies a stable embedded operating regime. In the calibrated fixed-span model with $\alpha = 2$ and span 0.25, the maximal observed stable bands are $\varepsilon_{\max} = 0.5$ for $(d, k) = (8, 1), (8, 2),$ and $(12, 2)$, and $\varepsilon_{\max} = 0.2$ for $(12, 1)$.

The same synthetic family also separates two mechanisms. Raw W_2 comparison between the triangular and i.i.d. empirical measures slows once intrinsic dimension increases, so the embedded fixed- ε statement does not follow from the available W_2 transfer route. Direct Jacobian probes of the centered Sinkhorn scaling map instead isolate a moderate-band regime: on $\varepsilon \in [0.2, 0.8]$, the cross-coupling block stays contractive, while the self-coupling blocks are the main source of instability but appear numerically to plateau with sample size.

The calibrated fixed-span model shows that the tested grid has a stable core and a small mixed boundary region. Calibration details and supporting tables are kept in appendix B. The strongest support appears on moderate bands bounded away from zero.

To probe whether the frontier is controlled more by local intrinsic geometry than by the ambient coordinate count, we estimate the effective intrinsic dimension with TwoNN on the same synthetic frontier family. On the nine-pair, six- ε grid studied here, the resulting $\hat{k}$ predicts the stable-band cutoff better than ambient dimension and improves the binary cut at $\varepsilon = 0.2$. This is consistent with geometry-aware calibration: the relevant control variable is the estimated local geometry, not the ambient coordinate count. TwoNN therefore enters here as a regime-selection diagnostic for the band analysis.

4.5 Temporal Validity Can Fail Before Detector Alarm

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over time. The horizon n_t^* contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\text{val-det}} := t_{\text{detect}} - t_{\text{valid}},$$

where t_{valid} is the first time the operating horizon persistently exceeds the temporal-validity horizon, and t_{detect} is the first detector alarm that follows.

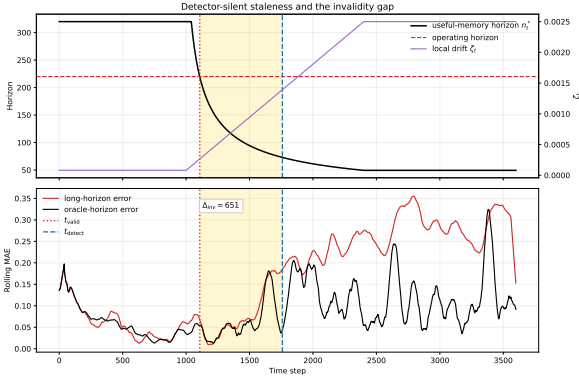


Figure 6: Temporal validity can fail before detector alarm. As drift intensifies, the temporal-validity horizon contracts until the operating horizon becomes stale at t_{valid} ; the detector responds only later at t_{detect} , leaving a strictly positive validity-detection lag. The lower panel shows excess tracking error accumulating before the alarm appears.

Across the reported runs, the measured lag remains positive. Retained evidence can already be invalid for the present while detector statistics remain below alarm threshold.

The positive lag is not a knife-edge feature of one tuning choice. Across the tested ADWIN thresholds, mean detection time remains strictly later than mean validity-expiry time, and the same qualitative lag persists under a different detector family as well. The empirical claim is robustness of the sign of $\Delta_{\text{val-det}}$ on the tested detector/tuning family, not a detector-uniform delay law.

4.6 Summary of Evidence

Taken together, the experiments support five claims that match the theory. Finite-memory tracking exhibits a structural U-curve with a near-optimal useful-memory region. The temporal-validity horizon is roughness dependent. The same retention pattern appears on the ELEC2 stream. The fixed- ϵ Sinkhorn frontier receives structured empirical support on the tested grid, with the strongest evidence concentrated on moderate bands bounded away from zero. Temporal validity can fail before change-point evidence appears, producing a measurable validity-detection lag.

5 Related Work

Nonstationary optimization and dynamic regret.

Dynamic-regret and variation-budget analyses show that under drift, forgetting scales and window lengths must adapt to how quickly the target moves [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. That literature isolates the control importance of retention scale. The present work supplies the corresponding statistical law for distribution tracking: finite memory has a temporal-validity horizon determined jointly by a finite-sample exponent and a drift roughness exponent.

Adaptive windowing, drift detection, and update policies.

Adaptive-window and concept-drift methods

study when accumulated evidence is sufficient to react [Bifet and Gavalda, 2007, Gama et al., 2004, 2014]. Hinder et al. [2023] emphasize that the window length itself is part of the problem. Resource-aware update scheduling under drift addresses a related decision problem in which update cost must be balanced against accumulated loss [Piaseczny et al., 2025]. Streaming OT and MMD detectors contribute the alarm-time perspective, emphasizing delay, power, and memory usage [Shvetsov et al., 2020, Kalinke et al., 2025]. These works provide the reaction timescale. The horizon law adds the validity timescale: retained evidence can cease to represent the present before change statistics cross an alarm threshold.

Adaptive filtering and locally stationary smoothing.

Classical adaptive filtering and state-space tracking tune forgetting factors, gain schedules, or process-noise levels to balance smoothing against responsiveness [Kalman, 1960, Jazwinski, 1970]. Locally stationary smoothing and bandwidth selection solve the analogous bias-variance problem for evolving regression or spectral targets [Dahlhaus, 1997]. These literatures supply mechanisms for adaptation. The present work contributes the distributional law that such mechanisms must respect when the retained object is an empirical distribution rather than a latent state or a local regression surface.

Distribution tracking under transport geometry.

Wasserstein distance and Sinkhorn-type estimators provide the finite-sample geometries that enter the horizon law. For empirical Wasserstein distances, Fournier and Guillin [2015], Boissard and Le Gouic [2014], and Weed and Bach [2019] show that the sampling exponent depends on geometric complexity even before drift enters. The horizon law takes that finite-sample exponent as one input and composes it with a staleness exponent induced by the drift path. In this sense, the transport literature provides the sampling side of the law, while the present work supplies its temporal counterpart.

Entropic and smoothed transport as operational geometries.

Regularized transport changes both statistical stability and computational behavior [Genevay et al., 2019, Mena and Niles-Weed, 2019, del Barrio et al., 2023, Rigollet and Stromme, 2025, Stromme, 2023, Groppe and Hundrieser, 2024, Eckstein and Nutz, 2022, Ghosal et al., 2022]. Fixed- ϵ Sinkhorn theory provides i.i.d. asymptotics, differentiability, null limit laws, and stability constants [Goldfeld et al., 2024]. Those results are the finite-sample anchor for the operational horizon law developed here. Recent linear-Gaussian entropic-OT analyses make the recursive structure of regularized transport explicit [Akyildiz et al., 2024], suggesting a constructive intermediate regime between fixed- ϵ i.i.d. asymptotics and fully drifted triangular-array tracking.

Intrinsic geometry and operational calibration.

Geometry-aware calibration of regularization scales points toward intrinsic dimension as the relevant complexity variable in regularized transport [Genens and Wintemberger, 2026]. Local intrinsic-dimension diagnostics such

as TwoNN provide a direct proxy for effective support geometry [Facco et al., 2017]. In the present work, that perspective enters through the operational Sinkhorn frontier: intrinsic geometry helps identify which finite-sample regime is available, and therefore which horizon law is relevant.

Low-dimensional proxy geometries. Sliced, max-sliced, subspace-robust, and projection-robust Wasserstein constructions replace ambient transport by one-dimensional or low-dimensional comparisons [Kolouri et al., 2018, Deshpande et al., 2019, Paty and Cuturi, 2019, Lin et al., 2020, Nadjahi et al., 2020, Nietert et al., 2022]. These examples show concretely that geometry choice can change the finite-sample exponent. They therefore fit naturally into the horizon framework as alternative sampling geometries whose drift-induced useful-memory scales need not match raw Wasserstein behavior.

Wasserstein nonstationarity and Age of Information. Wasserstein nonstationarity measures already appear in online optimization and distributionally robust optimization under drift [Jiang et al., 2020, Keehan et al., 2025, Amini and Vinas, 2026]. The Age-of-Information literature treats age as a resource that must be balanced against communication or update costs [Kaul et al., 2012, Yates et al., 2021]. Together these viewpoints make age and nonstationarity into operational variables. The horizon law gives them a distributional statistical interpretation: age is costly because it accumulates geometric mismatch with the present, and the roughness exponent determines how fast that cost grows.

Across these neighboring literatures, the shared theme is that nonstationarity makes retention scale matter. The present work identifies the statistical object that organizes that dependence for distribution tracking: a temporal-validity horizon, together with a useful-memory region and a distinct detectability timescale.

6 Limitations and Open Problems

The central structural object is the temporal-validity horizon for finite-memory tracking under drift, and the closed operational corollary is the induced useful-memory region. The theorem-level results are the general horizon law, the useful-memory region proposition, the uniform-window staleness bound, the one-dimensional root- n proof model, the structural Gaussian lower bound at the exponent level, and the Gaussian location minimax benchmark. The remaining claims are narrower.

Finite-sample scope. The theory is modular in the finite-sample exponent a , but the fully proved finite-sample instantiation established here is the one-dimensional root- n regime under bounded support and fixed span. The same balancing logic extends to other geometries only once an appropriate finite-sample theorem is supplied.

Operational scope. For fixed- ε Sinkhorn, the structural ingredients of the embedded fixed-span model are

proved, and the compact-support one-dimensional pointwise and bandwise inheritance theorems are explicit. The remaining operational step is the embedded low-dimensional theorem that connects the triangular-array finite-sample term to the i.i.d. benchmark uniformly across a nontrivial regularization band. The numerical evidence is strongest on moderate compact bands bounded away from zero, where direct Jacobian probes remain stable.

Lower-bound scope. The structural lower bound is subclass based. The Gaussian construction is enough to show that, in the root- n regime, the horizon is structural rather than an artifact of a particular estimator. Class-tightness and constant-sharpness for broader deterministic Hölder path classes remain outside the present closure.

Family-level scope. Regular dominated families are the natural extension class. The horizon logic is expected to carry over when local testing geometry and metric geometry remain comparable, but no theorem-level family law is proved here. Support-changing and singular families remain obstruction classes.

Weights and memory shape. Uniform windows are the main case. The open question is whether broader non-uniform weights can enlarge the useful-memory region or improve constants without changing the exponent across richer path classes.

Online roughness estimation. The law is stated in terms of the roughness parameters H and ζ , but no fully online procedure is proposed here for estimating them or for keeping the memory inside the useful region under misspecification.

Empirical scope. The experiments are structurally motivated. Synthetic designs make the horizon law, useful-memory region, and validity-detection lag visible under controlled drift, while the ELEC2 diagnostic checks that the same retention signature appears on a real stream. Their role is to expose the object, not to demonstrate competitive performance on a benchmark suite.

Detectability versus validity. Validity loss before detector alarm is an empirical consequence of the theory, not the primary theorem object. Temporal validity can fail before detector alarms appear, but it does not prove a universal detection-delay theorem across detector classes.

Open problems. Three directions enlarge the useful-memory theory directly: memory design, online adaptation, and validity-detection theory.

Problem 1: useful-memory design beyond uniform windows. The analysis is built around uniform windows, with only brief discussion of other weighting schemes. The open question is whether non-uniform weights can enlarge the useful-memory region or improve constants without changing the exponent across broad

path classes. A good answer would connect the horizon law to practical forgetting schemes without collapsing back into heuristic tuning.

Problem 2: online useful-memory adaptation.

The law depends on the roughness parameters H and ζ , but no online procedure is given here that estimates them and keeps memory inside the useful-memory region in real time. The open problem is to design an adaptive scheme with a provable excess-risk guarantee relative to the oracle region.

Problem 3: validity and detectability as coupled but distinct objects.

The experiments show a positive validity-detection lag, but there is no theorem yet relating exit from the useful-memory region to detector delay. The open problem is to prove detector-specific lower or upper laws that quantify when validity must fail before statistical detection.

7 Conclusion

Under drift, memory is never purely beneficial. It reduces finite-sample noise, but it also ages. The resulting trade-off makes finite memory a temporal-validity object with a finite horizon.

That idea is formalized through the horizon law

$$\mathbb{E} d\left(\hat{P}_t^{(n)}, P_t\right) \leq C_K n^{-a} + C_S \zeta n^H,$$

whose optimal scale is

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

Finite-memory tracking under drift is governed by this balance between finite-sample accuracy and drift-induced staleness, and the same balance induces a useful-memory region around the optimal scale.

A tractable one-dimensional proof model supplies a rigorous root- n finite-sample regime, and a structural Gaussian lower-bound construction shows that the corresponding horizon is not an artifact of one estimator. The same object yields a Gaussian location minimax benchmark on deterministic Hölder paths. In fixed- ε Sinkhorn geometry, the compact-support one-dimensional pointwise and bandwise inheritance theorems show that the same horizon logic remains visible in an operational transport geometry.

The finite-memory U-curve, roughness-dependent horizon scaling, a useful-memory region around the optimal scale, and a strictly positive lag between validity loss and detector alarm all fit the same horizon law. The resulting message is both theoretical and operational: retained evidence has a path-dependent horizon, useful memory occupies a finite near-optimal region, and statistical detectability need not coincide with temporal validity.

Temporal validity of retained evidence is therefore a first-class statistical object under drift. The horizon law identifies its governing scale, the useful-memory region records its operational range, and the detectability lag shows that validity can fail before alarms appear. Those three statements define the contribution of the paper.

Acknowledgements

With gratitude to everyone who contributed ideas, feedback, tools, encouragement, and care along the way. This work stands on the generosity of colleagues, collaborators, open-source maintainers, and the broader research communities that made it possible.

A Proof Details

This appendix records detailed derivations for structural results whose full calculations would interrupt the main line.

A.1 Gaussian lower-bound constants

Let

$$S_h := \sum_{r=1}^h r^{2H} \quad \text{and} \quad G_h := \sum_{r=1}^h (h^H - (h-r)^H)^2.$$

For the Gaussian two-point model with equal priors and common covariance $\sigma^2 I_h$, the Bayes classification error between means $\pm m$ is $\Phi(-\|m\|/\sigma)$. Therefore the ramp and endpoint-minimal payoffs are exactly

$$L_h^{\text{ramp}}(\sigma, \zeta) = \zeta h^H \Phi\left(-\frac{\zeta}{\sigma} \sqrt{S_h}\right)$$

and

$$L_h^{\text{min}}(\sigma, \zeta) = \zeta h^H \Phi\left(-\frac{\zeta}{\sigma} \sqrt{G_h}\right).$$

Proof of proposition 2.8. Fix $H \in (0, 1]$ and write $p_H = 2H/(2H+1)$.

For the ramp profile,

$$S_h \sim \frac{h^{2H+1}}{2H+1}.$$

Set

$$h = A \left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)}$$

with $A > 0$ fixed. Then

$$\frac{\zeta}{\sigma} \sqrt{S_h} \rightarrow \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

At the same time,

$$\zeta h^H = A^H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Hence

$$\frac{L_h^{\text{ramp}}(\sigma, \zeta)}{\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}} \rightarrow A^H \Phi\left(-\frac{A^{H+1/2}}{\sqrt{2H+1}}\right).$$

Define

$$x = \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

Then

$$A^H = (2H+1)^{H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$(2H+1)^{H/(2H+1)} x^{p_H} \Phi(-x).$$

Its logarithmic derivative is

$$\frac{p_H}{x} - \frac{\varphi(x)}{\Phi(-x)}.$$

Since the inverse Mills ratio $\varphi(x)/\Phi(-x)$ is strictly increasing on $(0, \infty)$, this objective has a unique maximizer $x_H > 0$, characterized by

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

This proves

$$\sup_{h \geq 1} L_h^{\text{ramp}}(\sigma, \zeta) \sim C_H^{\text{ramp}} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\text{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

For the endpoint-minimal profile, the Hölder constraint and endpoint condition imply

$$g_r \geq g_h - (h-r)^H = h^H - (h-r)^H$$

for every feasible endpoint-saturating profile g . Therefore the profile

$$g_r^{\min} = h^H - (h-r)^H$$

minimizes testing energy pointwise and hence minimizes $\sum_{r=1}^h g_r^2$. Its rescaled energy satisfies the Riemann-sum limit. Thus

$$h^{-(2H+1)} G_h \rightarrow I_H,$$

where

$$I_H = \int_0^1 (1 - (1-u)^H)^2 du = \frac{2H^2}{(H+1)(2H+1)}.$$

With the same scaling

$$h = A \left(\frac{\sigma}{\zeta} \right)^{2/(2H+1)},$$

one has

$$\frac{\zeta}{\sigma} \sqrt{G_h} \rightarrow A^{H+1/2} \sqrt{I_H}.$$

Setting $x = A^{H+1/2} \sqrt{I_H}$ gives

$$A^H = I_H^{-H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$I_H^{-H/(2H+1)} x^{p_H} \Phi(-x).$$

The same maximizer x_H therefore yields

$$\sup_{h \geq 1} L_h^{\min}(\sigma, \zeta) \sim C_H^{\min} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Finally,

$$\frac{C_H^{\min}}{C_H^{\text{ramp}}} = \left(\frac{I_H^{-1}}{2H+1} \right)^{H/(2H+1)} = \left(\frac{H+1}{2H^2} \right)^{H/(2H+1)}.$$

This factor is strictly larger than 1 for $H < 1$ and equal to 1 at $H = 1$. $\square$

A.2 Two-point lower bound used in proposition 2.7

The structural lower bound above is the exponent-level consequence of the exact Gaussian two-point calculation. For the ramp pair

$$\mu_{t-j}^{\pm, h} = \pm \zeta (h-j)_+^H,$$

the present-time separation is $2\zeta h^H$, so any estimator incurs risk at least of order $L_h^{\text{ramp}}(\sigma, \zeta)$. Minimizing over h and using the asymptotic just proved gives the lower scale

$$\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

and therefore the horizon exponent $(2H+1)^{-1}$ for the root- n regime.

B Finite-Sample Diagnostics

This appendix collects finite-sample diagnostics that support the theorem line and the conjectural extensions. These checks are not substitutes for theorem statements.

B.1 Finite-Sample Geometry Check

For the finite-sample term, we estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log-log slope in n . We also probe the fixed-span one-dimensional triangular window directly by comparing a triangular-array sample, with one draw from each drifted component, against an i.i.d. sample from the same window mixture. In that check the within-window span ζn^H is held fixed. Finally, we check the Sinkhorn proxy at a few fixed values of ε .

Table 1: Empirical scaling check for the finite-sample term. The bounded-support i.i.d. W_2 experiment is close to the expected $n^{-1/2}$ rate in low dimension, the corresponding triangular-array check compares one sample from each drifted component against an i.i.d. sample from the window mixture, and the Sinkhorn proxy is reported at fixed ε . These rows are numerical support for the carrier-regime discussion rather than theorem statements.

Setting	Estimated slope	Comment
$W_2, d = 1$	-0.48	close to $-1/2$
$W_2, d = 2$	-0.41	slower decay
$W_2, d = 4$	-0.25	slower decay
$W_2, d = 5$	-0.24	slower decay
$W_2, \text{i.i.d. mixture}, d = 1, H = 0.5$	-0.50	fixed span $\zeta n^H = 0.25$
$W_2, \text{triangular}, d = 1, H = 0.5$	-0.45	fixed span $\zeta n^H = 0.25$
$W_2, \text{i.i.d. mixture}, d = 1, H = 1.0$	-0.47	fixed span $\zeta n^H = 0.25$
$W_2, \text{triangular}, d = 1, H = 1.0$	-0.41	fixed span $\zeta n^H = 0.25$
Sinkhorn $\varepsilon = 0.50$	-0.21	proxy constant changes
Sinkhorn $\varepsilon = 0.10$	-0.53	proxy constant changes
Sinkhorn $\varepsilon = 0.05$	-0.67	proxy constant changes

This table supports the finite-sample interpretation used above. In low dimension, the raw i.i.d. Wasserstein term is compatible with a root- n rate, whereas higher-dimensional raw Wasserstein slows down. In the one-dimensional fixed-span triangular-array check, the recovered slopes stay near the corresponding i.i.d. mixture baseline, which is consistent with the proof-model and design-effect reading used in the 1D proof-model proposition.

B.2 Extended and Operational Regimes

The next table summarizes broader finite-sample evidence beyond the one-dimensional proof model. The extended rows compare triangular and i.i.d. mixture rates on embedded low-intrinsic-dimensional support across span. The operational rows compare fixed- ϵ Sinkhorn rates across regularization values on embedded low-intrinsic-dimensional support in ambient dimension eight.

Table 2: Representative rows for the extended and operational finite-sample regimes. The extended rows compare triangular and i.i.d. mixture rates across span, while the operational rows compare fixed- ϵ Sinkhorn across ϵ on embedded low-intrinsic-dimensional support.

Regime	Setting	Parameter	Rate exponent α
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.10	0.10	0.4652
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.10	0.10	0.5722
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.25	0.25	0.4737
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.25	0.25	0.5588
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.50	0.50	0.4792
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.50	0.50	0.5425
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, $\epsilon=0.50$	0.50	0.5599
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, $\epsilon=0.50$	0.50	0.5706
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, $\epsilon=0.20$	0.20	0.5466
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, $\epsilon=0.20$	0.20	0.6150
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, $\epsilon=0.10$	0.10	0.4646
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, $\epsilon=0.10$	0.10	0.5060
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, $\epsilon=0.05$	0.05	0.4339
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, $\epsilon=0.05$	0.05	0.5124

These rows provide calibration for the theorem statements above. The extended-regime rows support the claim that low intrinsic dimension preserves a rate near the root- n regime without a dramatic triangular-versus-benchmark gap. The operational-regime rows supply empirical evidence relevant to the fixed- ϵ Sinkhorn setting in definition 2.10: together with exact support-complexity inheritance and the dual threshold $\alpha > k/2$, they support the conjectural inheritance statement discussed in conjecture 2.14. A full proof of triangular-array inheritance would remove this empirical calibration step.

Two complementary diagnostics sharpen the same boundary. First, direct probes of the centered Sinkhorn scaling Jacobian on the embedded fixed-span model show that bands approaching zero are the numerically unstable region, whereas on a moderate band such as $[0.2, 0.8]$ the cross-coupling block remains contractive and the self-coupling inverse norms appear to plateau with sample size. Second, on the supported synthetic frontier grid, TwoNN-based intrinsic-dimension estimates predict the observed stable-band cutoff better than ambient dimension. These diagnostics identify a useful regime proxy and isolate the band where the remaining instability is concentrated. They do not upgrade the embedded fixed- ϵ statement from conjecture to theorem.

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